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1 Materials and Methods

1.1 Effective population size trajectories and feature extraction

Posterior trajectories of the effective population size were taken from Bayesian Skygrid coalescent analyses (Gill et al. 2013) performed in BEAST (v1.10) (Suchard et al. 2018), with 200 posterior draws retained for each of the eight geographic categories after the first 10 % of each chain had been discarded as burn-in and the remaining states had been thinned to at most 4,000 draws. All draws were interpolated onto a common calendar grid running from the present to 20,000 years before present in steps of 200 years, the upper bound being the oldest time covered by every input. Each draw was smoothed and differentiated in a single pass by locally weighted quadratic regression with a tricube kernel and a bandwidth of 3,000 years (Cleveland and Devlin 1988), yielding a smoothed $\log N_e$ curve and its rate of change expressed per 1,000 years. The time axis was divided into 40 non-overlapping windows of 500 years, and eight trajectory features (mean level, net change, mean rate of change, fluctuation amplitude, and the timing and magnitude of the strongest expansion and contraction) were computed for every draw within each window; medians, 95 % highest posterior density (HPD) intervals and directional posterior probabilities were then summarised across draws. A grid point was regarded as reliable when the 95 % HPD width of $\log N_e$ did not exceed 2.0 log units, and a category entered a given window only when at least 80 % of the grid points of that window were reliable. All computations were implemented in Python (v3.10.20) with NumPy (v2.2.6) (Harris et al. 2020), SciPy (v1.15.2) (Virtanen et al. 2020) and pandas (v2.3.3) (McKinney 2010).

1.2 Trajectory clustering, ordination and visualisation

Within each time window the eight features were standardised across categories (z-scores) and clustered by average-linkage hierarchical clustering of Euclidean distances using SciPy (v1.15.2) (Virtanen et al. 2020); the number of clusters was selected between $K = 2$ and $K = 8$ by maximising the average silhouette coefficient (Rousseeuw 1987) computed with scikit-learn (v1.7.2) (Pedregosa et al. 2011). Cluster labels were made comparable between consecutive windows by optimal label matching with the Hungarian algorithm (Kuhn 1955). Uncertainty in the partition was quantified by 500 posterior resamples in which one curve per category was drawn at random, the features were recomputed and the clustering repeated with the same K (random seed 20260828); the proportion of resamples placing two categories in the same cluster was reported as the co-clustering frequency, and pairs with a frequency of at least 0.5 were drawn as chords. Both the co-clustering frequency and a feature similarity, defined as one minus the Euclidean distance normalised by its within-window maximum, were additionally displayed as category-by-category matrices, with rows and columns kept in a single global order across all windows. The scaled feature matrix of each window was further projected onto its first two principal components (Pedregosa et al. 2011), and two composite scores (a level score from mean level and amplitude, and a change score from net change, rate and the extreme expansion and contraction rates) were defined as the first principal component of the corresponding feature subsets and standardised within each 500 years bin. Figures were produced in R (v4.6.0) (R Core Team 2026) with TIDYPLOTS (v0.3.1) (Engler 2025), GGPLOT2 (v4.0.3) (Wickham 2016), GGREPEL (v0.9.8) (Słowikowski 2025), PHEATMAP (v1.0.13) (Kolde 2025), and IGRAPH (v2.3.3) (Csárdi and Nepusz 2006) together with GGRAPH (v2.2.2) (Pedersen 2024).

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